Fiscal Year:	FY 2020	**Task Last Updated:**	FY 07/06/2023
PI Name:	Simpson, Richard Ph.D.		
Project Title:	The Impact of Simulated Microgravity on the Anti-Tumor Properties of Human NK-Cells and ? d -T Cells In Vivo: IL-2 and Zoledronic Acid Therapy as a Potential Countermeasure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0001	**Congressional District:**	3
Comments:	NOTE: Formerly at University of Houston until September 2017 move to University of Arizona.		
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	05/21/2019	**End Date:**	05/20/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Katsanis, Emmanuel M.D. (The University of Arizona)		
Grant/Contract No.:	80NSSC19K1059		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Immune system impairment has been documented during and after long-duration spaceflight, which could increase the risk of cancer and latent viral reactivation in exploration class mission crew. Natural Killer (NK)-cells are a critical component of the immune system and play a vital role in anti-tumor surveillance and viral control. We have shown that the anti-tumor activity of NK-cells is impaired in International Space Station (ISS) crewmembers during the flight phase of the mission and is also compromised when normal human NK-cells are exposed to simulated microgravity in vitro. As such, it is critical to determine the effects of microgravity on the anti-tumor activities of NK-cells in vivo to determine if this poses a serious risk to the health and safety of exploration class mission crew. Moreover, it is equally important to start validating potential immune system countermeasures in preparation for long duration missions. The main goal of this project is to determine the effects of simulated microgravity on the anti-tumor activity of human NK-cells and γ δ -T cells in vivo using a humanized experimental mouse model. We will also test the efficacy of a potential countermeasure that is currently used in allogeneic and autologous stem cell transplant settings to boost NK-cell and γ δ -T cell function in vivo after transplant. This goal will be achieved through the following specific aims: SA1: Determine the effect of simulated microgravity on the anti-tumor capacity of human NK-cells and γ δ -T cells in vivo. Experimental Approach: We will use our established rotating cell culture model to expose human peripheral blood mononuclear cells (PBMCs), expanded NK-cells or expanded γ δ -T cells to simulated microgravity or controls (static-1G and rotational control) prior to adoptive transfer into human tumor bearing NOD-scid-IL2R γ null (NSG) mice. The NSG mice will be engrafted with a human luciferase-labelled human leukemia cell line (K562), as blood cancers are considered a risk for long duration exploration crew. The NSG mice will also be engrafted with an NK-cell sensitive solid tumor (human neuroblastoma: SK-N-SH cell line) which will enable us to study tumor infiltrates by isolating tissues and quantifying the human immune cells in the tumor by multi-parameter flow cytometry. Spectral imaging (LagoX) will be utilized to measure tumor progression and dissemination of the luciferase-tagged human cancer cell lines. Hypothesis: Exposure to simulated microgravity will have an adverse effect on the in vivo dynamics of human NK and γ δ -T cells, impairing their ability to infiltrate and control human tumor growth in a humanized rodent model. SA2: Determine the effect of systemic administration of Interleukin-2 (IL-2) and zoledronic acid (ZOL) on the function of NK and γ δ -T cells respectively. Experimental Approach: NSG mice adoptively transferred with human PBMCs, NK-cells and γ δ -T cells will be infused with ZOL and IL-2 as a potential countermeasure to microgravity-induced immune system impairment. ZOL and IL-2 are currently used clinically to increase the proliferation and function of NK-cells and γ δ -T cells in bone marrow transplant patients to prevent disease relapse. Hypothesis: Systemic administration of IL-2 and ZOL will 'rescue' the in vivo anti-tumor activity of human NK-cells and γ δ -T cells following their exposure to simulated microgravity. By the end of this project we will have a better understanding on the effects of simulated microgravity on anti-tumor immunity in vivo. We will also make the first steps to determine the efficacy of an immune system countermeasure currently used in a clinical setting to boost the anti-tumor and anti-viral activity of human NK-cells and γ δ -T cell in vivo, which could have application for long-duration exploration class mission crew.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Update (Ed., 7/6/23) Results: Mice that received peripheral blood mononuclear cells (PBMCs) exposed to simulated microgravity (SMG) showed poorer control of tumor growth compared to mice that received PBMCs exposed to 1G. This suppression of in vivo function was evidenced by increased tumor growth rate (photons/sec) and peak tumor growth (maximum biolayer interferometry/BLI) attained throughout the duration of the experiment (40 days). Administration of systemic immuno-therapeutics: IL-2 (subcutaneous) and zoledronic acid (intraperitoneal) rescued this loss of function in vivo. Conclusion: Microgravity plays a contributory role in immune dysregulation observed in spaceflight. Future long duration exploration missions should consider effects of microgravity and partial gravity on immune health of the astronauts. Various prophylactic immune-therapeutic regimens might offer protective cover against various hazards of spaceflight.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Mylabathula PL, Diak DM, Baker FL, Niemiro GM, Markofski MM, Crucian BE, Katsanis E, Simpson RJ. "IL-2 and zoledronic acid therapy restores the in vivo anti-leukemic activity of human lymphocytes pre-exposed to simulated microgravity." Front Biosci (Landmark Ed). 2022 Jul 8;27(7):215. Review. https://doi.org/10.31083/j.fbl2707215 ; PMID: 35866402 , Jul-2022
Articles in Peer-reviewed Journals	Mylabathula PL, Li L, Bigley AB, Markofski MM, Crucian BE, Mehta SK, Pierson DL, Laughlin MS, Rezvani K, Simpson RJ. "Simulated microgravity disarms human NK-cells and inhibits anti-tumor cytotoxicity in vitro." Acta Astronaut. 2020 Sep;174:32-40. https://doi.org/10.1016/j.actaastro.2020.03.023 , Sep-2020